

2022 Ph H2 Q4

A spacecraft of mass 870 kg docks with a stationary satellite of mass 540 kg.

After docking, the two objects move together.

(a)

Calculate the velocity of the spacecraft and satellite immediately after docking

This is a **completely inelastic collision** — momentum is conserved, but kinetic energy is not.

Use conservation of momentum:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

Where:

- $m_1 = 870\text{ kg}$, $u_1 = 1.6\text{ m/s}$
- $m_2 = 540\text{ kg}$, $u_2 = 0$

$$(870)(1.6) + (540)(0) = (870 + 540)v$$
$$1392 = 1410v \Rightarrow v = \frac{1392}{1410} = 0.987\text{ m/s}$$

(b)

Calculate the total kinetic energy lost during the docking process

Initial kinetic energy:

$$E_{k,i} = \frac{1}{2}m_1u_1^2 = \frac{1}{2}(870)(1.6)^2 = \frac{1}{2}(870)(2.56) = 1113.6\text{ J}$$

Final kinetic energy (combined mass moving at 0.987 m/s):

$$E_{k,f} = \frac{1}{2}(1410)(0.987)^2 = \frac{1}{2}(1410)(0.974) = 686.4\text{ J}$$

Kinetic energy lost:

$$\Delta E_k = 1113.6 - 686.4 = 427\text{ J}$$

 Revision Tips

- In **inelastic collisions**, momentum is always conserved, but **kinetic energy is not**
- Always use **separate equations** for momentum and kinetic energy
- Watch out for **units** and ensure both masses and velocities are clearly defined before calculating
- In exam explanations, use precise terms like “momentum is conserved” and “kinetic energy is transformed into other forms (e.g., heat, sound, deformation)”

